

Progress Report - August 4, 2026

Lag and compression between satellite GVF and PhenoCam (NDVI / GCC)

Why these metrics

Divergence scores say how much the curves disagree overall. Lag and compression split that into two readable pieces: (1) whether GVF phase dates sit earlier or later than ground NDVI, and (2) whether the GVF greenup / senescence interval is shorter or longer than PhenoCam GCC.

How lag is measured

For each phase date (SOS, MOS, DOS, EOS), lag is the day-of-year difference GVF minus NDVI:

```
lag_sos = gvf_sos - ndvi_sos
lag_mos = gvf_mos - ndvi_mos
lag_dos = gvf_dos - ndvi_dos
lag_eos = gvf_eos - ndvi_eos
```

```
# lag (alias used in ranking) = lag_sos
```

Read as:

- lag > 0 : GVF phase is later than NDVI
- lag = 0 : same DOY
- lag < 0 : GVF phase is earlier than NDVI

Units are days. Typical |lag| ~0-15 d; 15-40 d is notable; 40+ d is extreme (also check spin-up: gvf_sos == 1 is a failed fit, not a real Jan 1 greenup).

How compression is measured

Compression compares the length of the GVF transition to the GCC transition (ratio, not days):

```
greenup_comp = (gvf_mos - gvf_sos) / (gcc_mos - gcc_sos)
senescence_comp = (gvf_eos - gvf_dos) / (gcc_eos - gcc_dos)
```

Read as (ratio form, used in single-year plots):

- ratio > 1 : GVF interval longer than GCC (stretched)
- ratio = 1 : same length
- ratio < 1 : GVF interval shorter than GCC (compressed)

Combined 2023/2024 compression figures plot (ratio - 1) so the axis is centered at 0 (same length). Positive = stretched; negative = compressed.

Implemented in shared/data_collection.py (_add_lag_compression). Plots from anomaly_pipeline/anomaly_check.ipynb -> anomaly_pipeline/output/lollipopPlot/.

How to read the plots

Lollipops: stem from 0 (or ratio=1) to the site value. Combined plots: blue = 2023, green = 2024; same site paired when both years exist. Veg focus: AG / DB / GR / SH. Figures 1-4 on the next pages.

GBOV + GoldenSites – Phase lag 2023 (blue) vs 2024 (green)

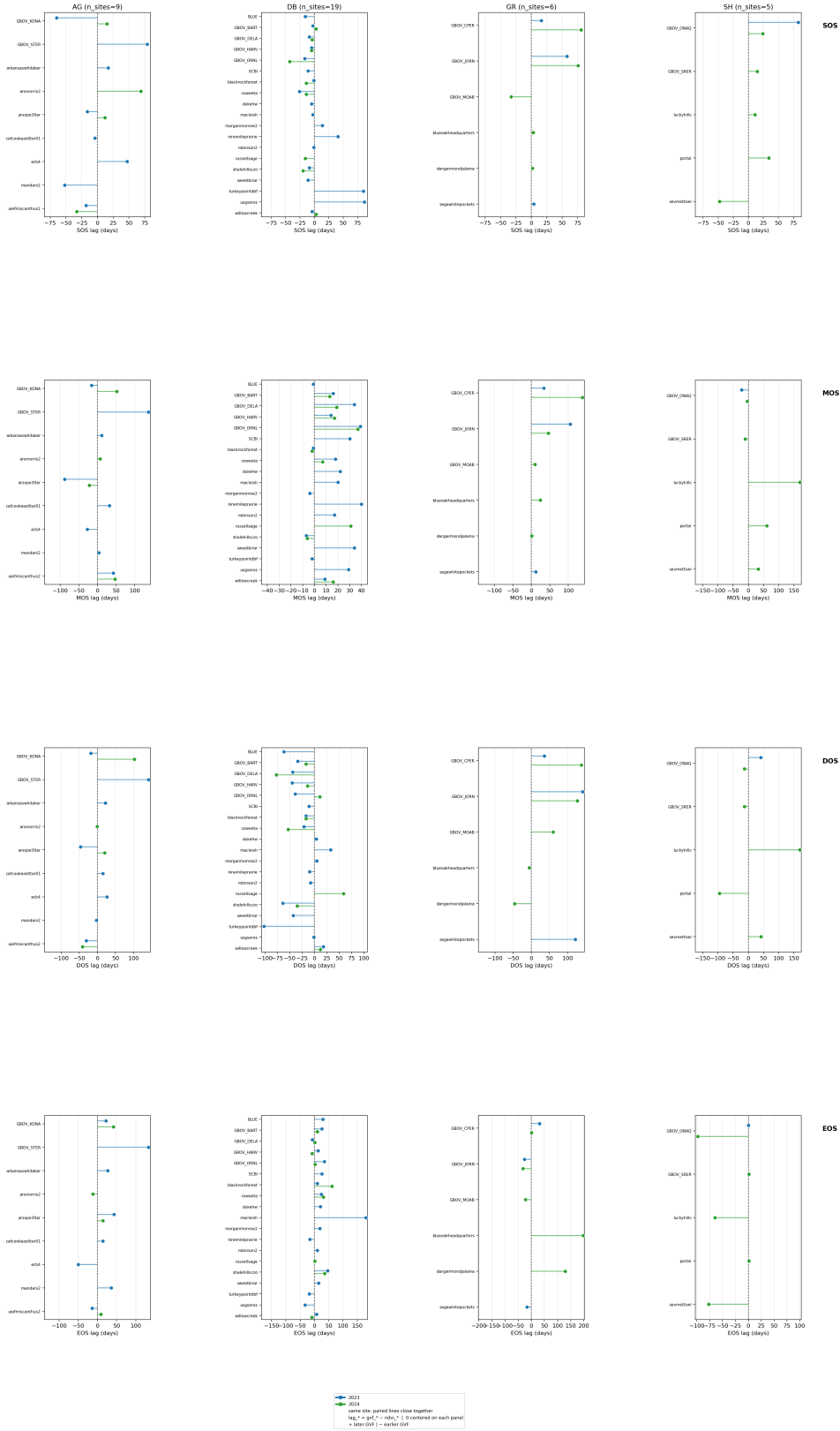


Figure 1. Combined phase lag (GBOV + GoldenSites), 2023 (blue) vs 2024 (green). Rows = SOS / MOS / DOS / EOS; columns = AG / DB / GR / SH.

GBOV + GoldenSites — Compression 2023 (blue) vs 2024 (green)

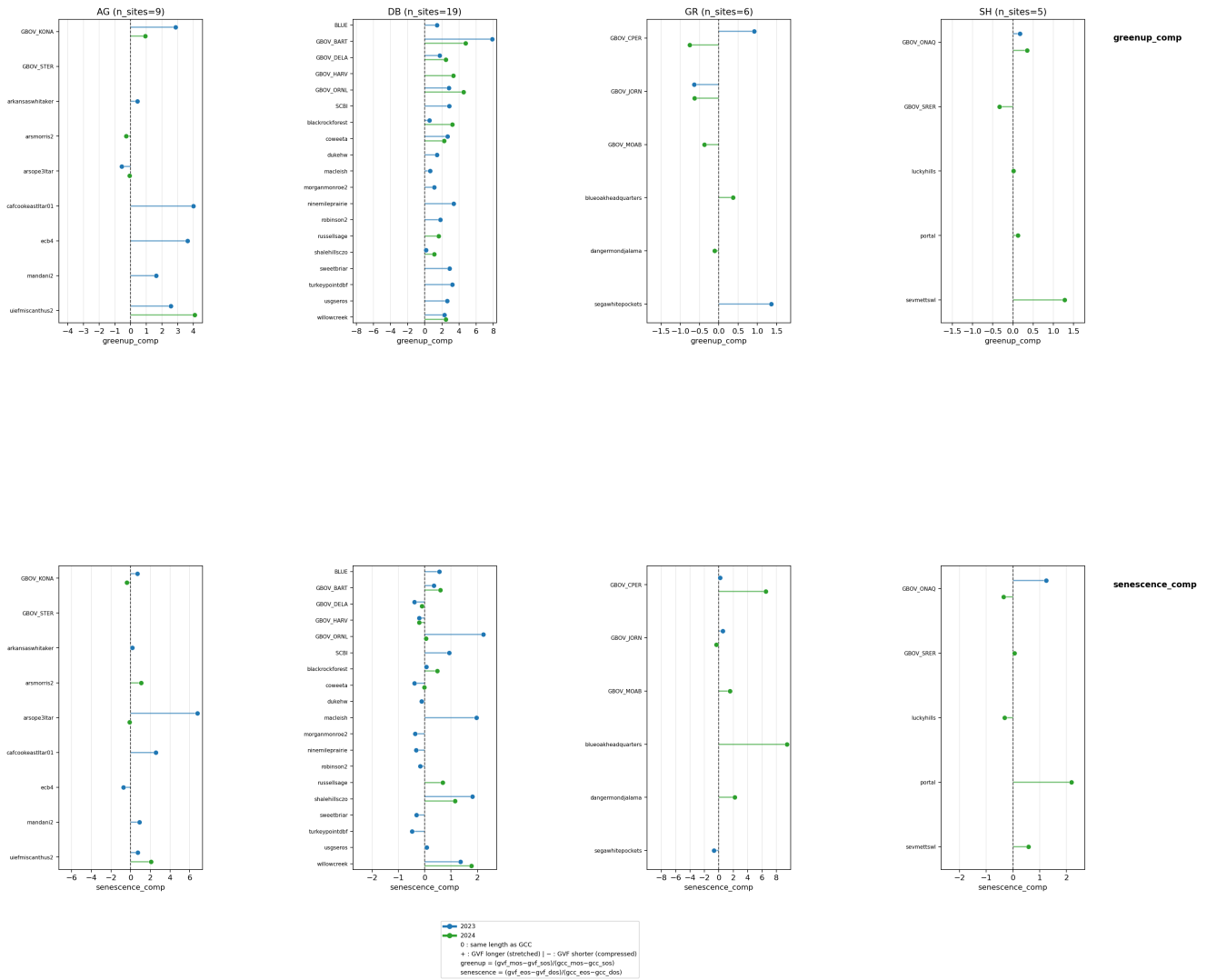


Figure 2. Combined compression (GBOV + GoldenSites), 2023 (blue) vs 2024 (green). Plotted as (ratio - 1) so 0 = same length as GCC; + = stretched, - = compressed.

2023 GoldenSites — Phase lag (GVF – NDVI) by veg

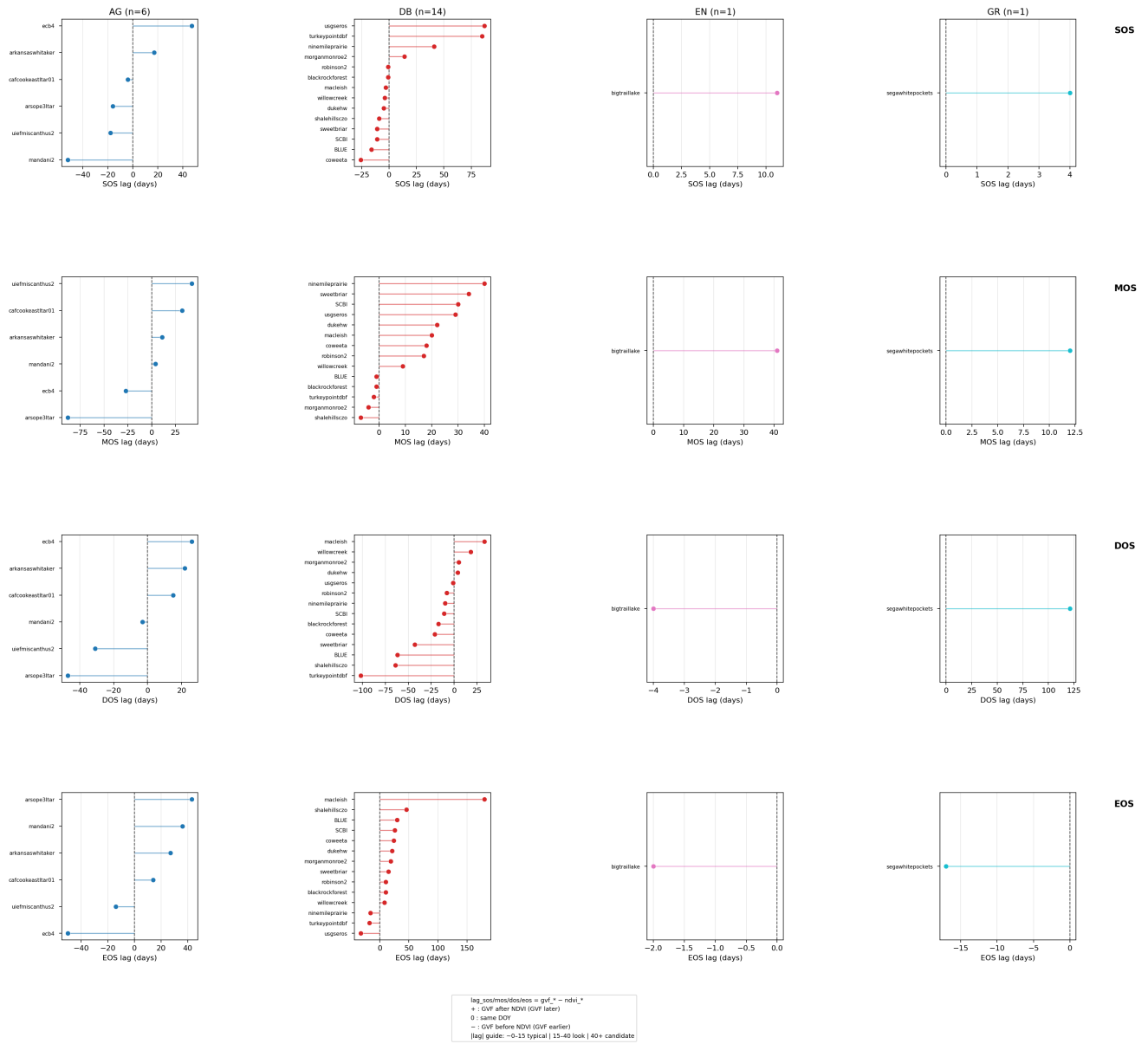


Figure 3. 2023 GoldenSites phase lag by veg (raw days). AG and DB dominate the sample.

2023 GoldenSites — greenup_comp (left) | senescence_comp (right)

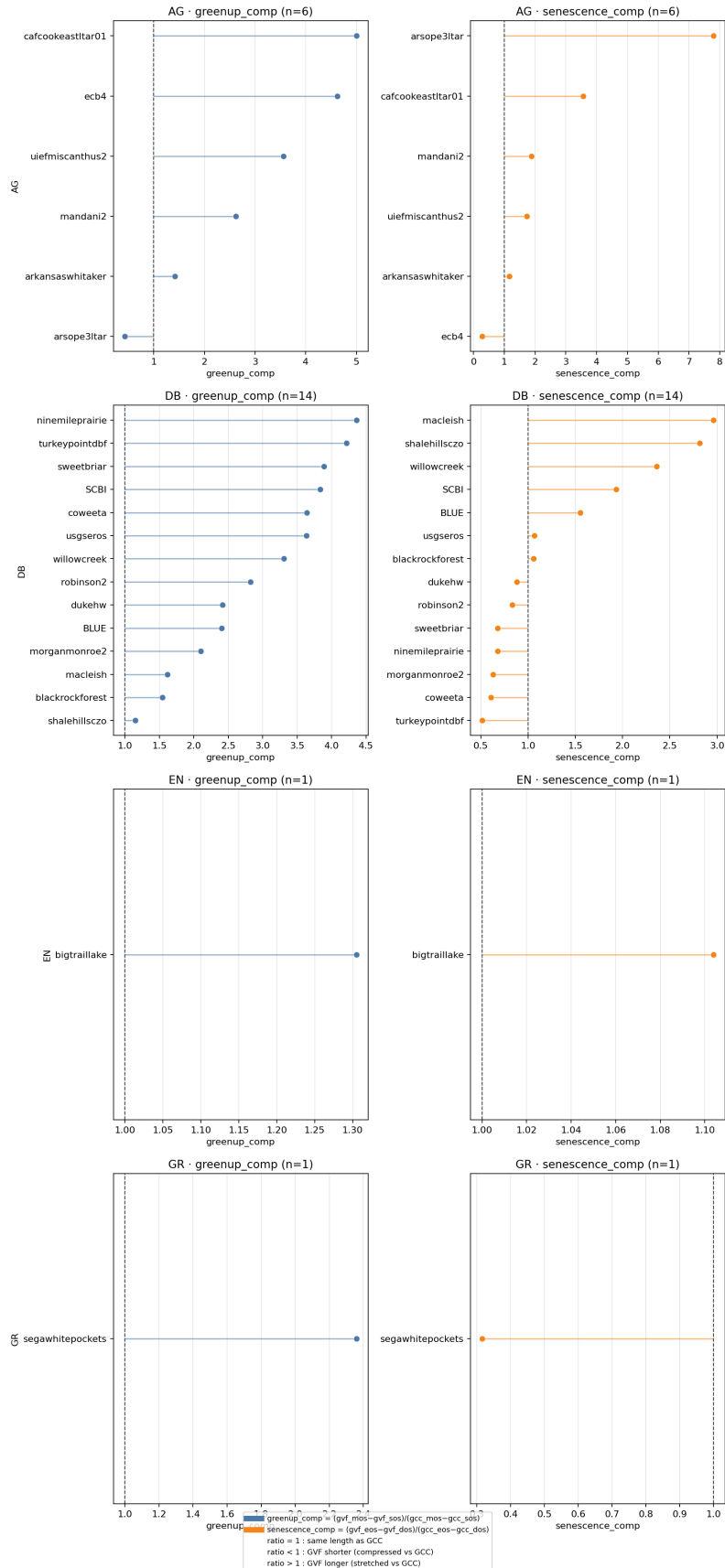


Figure 4. 2023 GoldenSites greenup / senescence compression by veg (ratio scale; 1 = match GCC).

Findings

GoldenSites 2023 (from anomaly_check notes)

- **Compression** - AG (n=6): greenup and senescence mostly > 1 (GVF stretched vs GCC). DB (n=14): greenup all stretched; senescence mixed (about half stretched / half compressed; a few near 1). EN and GR (n=1 each): both phases stretched.
- **Lag** - EN / GR: small samples, lag looks minor. AG: SOS lag spans roughly -50 to +50 days. DB: many sites a few days earlier or near zero, but a few ~+50 d later at SOS; DOS/EOS can show much larger offsets.

Combined GBOV + GoldenSites (2023 vs 2024)

- DB has the largest n and often clusters nearer zero at SOS/MOS than AG/GR/SH; DOS/EOS still show large outliers (100+ days) at some sites.
- Year-to-year: many AG/DB sites look less stretched in 2024 than 2023 on the combined compression plot (green closer to 0 than blue), but this is not universal.
- SH tends to sit closer to matched duration (near 0 on ratio-1) than AG/DB greenup, though lag at MOS/DOS can still be large for individual sites.

Caveats / next

- Exclude spin-up (gvf_sos == 1) before interpreting extremes.
- Lag uses NDVI phase dates; compression uses GCC interval lengths - they answer different questions and can disagree at the same site.
- Next: tie large lag/compression sites back to curve overlays and surface fractions (water% / urban%), and compare NOAA 1 km/4 km GVF at the same points when ready.